



SIMATS
ENGINEERING



SIMATS
Saveetha Institute of Medical And Technical Sciences
(Declared as Deemed to be University under Section 3 of UGC Act 1956)

Department of Artificial Intelligence and Data Science

CSA1407-COMPILER DESIGN

Role of Lexical Analyzer

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Introduction

Role of Lexical Analyzer

- The lexical analyzer is the first phase of a compiler.
- It reads the source program character by character.
- It converts the program into meaningful tokens.
- It removes unwanted characters like spaces and comments.
- It sends tokens to the syntax analyzer for further processing.

Scans Source Code

Scanning Source Code

- Reads the source code from beginning to end.
- Identifies characters, words, and symbols.
- Recognizes patterns using regular expressions.
- Prepares the input for token generation.
- Ensures every part of the program is scanned correctly.

Converts into Tokens

Token Generation

- Breaks the source code into small units called tokens.
- Tokens include keywords, identifiers, operators, constants, and separators.
- Each token has a specific meaning.
- Makes the program easier for the compiler to process.
- Improves the speed of compilation.

Removes Unnecessary Characters & Passes Tokens to Parser

Processing the Input

- Removes spaces, tabs, blank lines, and comments.
- Ignores unnecessary characters that do not affect execution.
- Sends the generated tokens to the syntax analyzer (parser).
- Helps the parser check the grammar of the program.
- Supports smooth communication between compiler phases.

Handles Lexical Errors

Lexical Error Handling

- Detects invalid characters and unknown symbols.
- Reports incorrect or invalid tokens.
- Displays error messages with the error location.
- Prevents invalid input from reaching the parser.
- Improves the reliability of the compilation process.

Conclusion

- The lexical analyzer is the first and important phase of a compiler.
- It scans the source code and converts it into tokens.
- It removes unnecessary characters and detects lexical errors.
- It passes valid tokens to the parser for syntax analysis.
- It makes the compilation process faster, easier, and more efficient.

THANK YOU!



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